
Supporting Information for

Recovery of iron resources from wet-process phosphoric acid and the preparation of high purity ferrous oxalate: Solvent extraction and photochemical reaction

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Page S3: Material and electrical energy consumption of producing 1 ton of $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (**Table S1**).

The main cost of producing $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ by photochemical reaction is the consumptions of oxalic acid and electric energy. According the equation (1) in the manuscript, 1.5 mol of $\text{H}_2\text{C}_2\text{O}_4$ is consumed to produce 1 mol of ferrous oxalate. The consumption of electrical energy is obtained through a continuous photochemical reaction experiment.

To investigate the consumption of electrical energy, a continuous photochemical reaction experiment was carried out with the experimental setup shown in **Figure S1**. The reactor volume is 1.8 L, the $\text{H}_2\text{C}_2\text{O}_4$ concentration = 1.20 mol/L, Fe(III) concentration = 0.06 mol/L, pH = 0.5. The actual power of the UV lamp was 395 nm, 90 W. The experiment was divided into three stages, and the first stage was an intermittent reaction: the reactor was filled with oxalic acid solution containing Fe(III) and the UV lamp was turned on. At the same time, a peristaltic pump was launched to continuously inject the solution from the upper end of the reactor to the lower end. The reaction time was 16 minutes. In the second and third stages, the reaction material was continuously fed and discharged at flow rates of 120 ml/min and 150 min/L, respectively.

The electrical energy consumption of the intermittent reaction is calculated as follows:

$$\text{Electrical energy consumption} = \frac{P \cdot t}{m} \quad (1)$$

The electrical energy consumption of the continuous reaction is calculated as follows:

$$\text{Electrical energy consumption} = \frac{P \cdot \Delta t}{\Delta m} \quad (2)$$

where P is the actual power of the UV lamp, kW; t is the cumulative light time, Δt is the light time in a certain time period, h; m is the cumulative amount of ferrous oxalate generated; Δm is the mass of ferrous oxalate generated in Δt time.

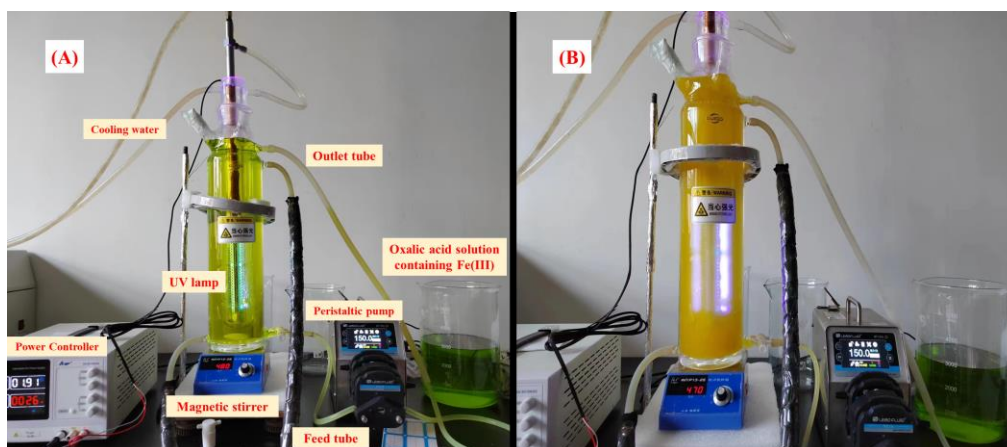


Figure S1. Experimental setup for continuous photochemical reaction, (A) before reaction, (B) after reaction.

The experiment results were shown in **Figure S2**. In the first stage, the total Fe content in the oxalic acid solution decreased sharply after 5 min, and the precipitation rate reached 92.09% at 13 min. The electric energy consumption decreased to a minimum of 1163 kW•h/t. When the second stage reached stability at 25 min, the precipitation rate and electric energy consumption were 88.97% and 1389 kW•h/t, respectively. When the third stage reached stability at 43 min, the precipitation rate and electric energy consumption were 77.90% and 1277 kW•h/t, respectively. Although the energy consumption was reduced with increasing flow rate, the precipitation rate decreased significantly. Therefore, a flow rate of 120 ml/min is suitable when the UV lamp power is 90 W.

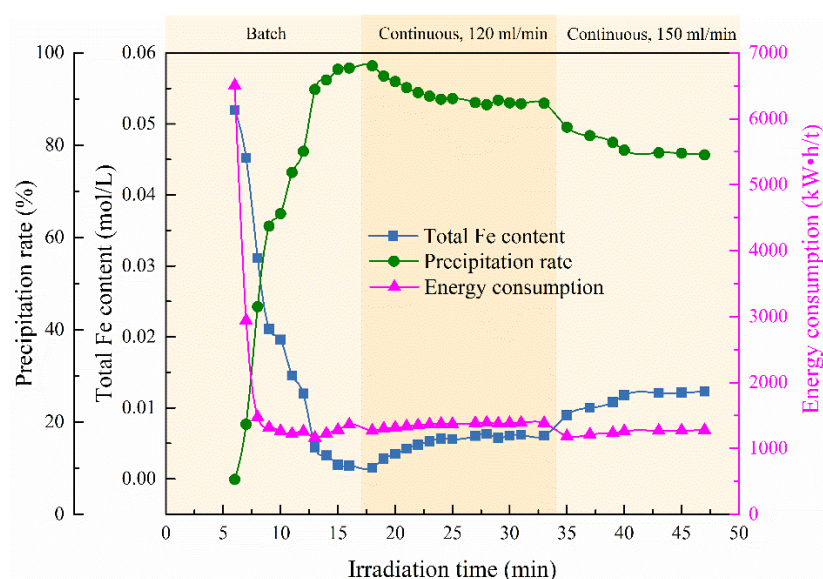


Figure S2. Electrical energy consumption of the production of $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ by photochemical

reactions.

Table S1 displayed the material and electrical energy consumption of producing 1 ton of $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$. Compared to the commercialized $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$, the method of this paper produces ferrous oxalate at a lower total cost of materials and electrical energy.

Table S1. Material and electrical energy consumption of producing 1 ton of $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$.

Name	Purity	Unit	Unit price/\$	Consumption	
				This work	Commercialized
$\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$	> 99%	ton	560	1.05	0.70
$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	> 99%	ton	378	-	1.54
Electric energy	-	$10^3 \text{ kW} \cdot \text{h}$	84	1.389	-
Total cost/\$	-	-	-	704.7	974.1